

Roll No.

TPE-724

**B. TECH. (PETROLEUM ENGINEERING)
(SEVENTH SEMESTER)**

MID SEMESTER EXAMINATION, Oct., 2022

PETROLEUM ECONOMICS, RISK AND UNCERTAINTY ANALYSIS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks

1. (a) What is petroleum economics ? Why is petroleum economics important in oil and gas industry ? When to do petroleum economics ? (CO1)

OR

- (b) Project feasibility study is an important part of the petroleum economics and risk analysis in oil and gas industry, discuss the studies to be conducted in oil and gas project feasibility study before the implementation of the project. (CO1)

2. (a) Discuss the life cycle of an oil and gas project. (CO2)

OR

- (b) What is infill drilling in oilfield ? Discuss why the analysis of infilling drilling is important for making oil and gas project more profitable and minimizing the risk. (CO2)

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(2)

3. (a) What is Form-in and Form-out agreements in oil and gas industry ?

(CO1)

OR

- (b) Explain risk and discuss the importance of risk analysis in oil and gas projects.

(CO1)

4. (a) What is cash flow statement ? What are the objectives of cash flow statement ?

(CO2)

OR

- (b) Discuss how the cash flow statement benefits the oil and gas industry projects.

(CO2)

5. (a) What is CAPEX and OPEX ? Explain how to calculate capital expenditures (CAPEX) for a project.

(CO2)

OR

- (b) What is abandonment cost in an oil and gas project ? Discuss the key well abandonment and decommissioning challenges faced by the oil and gas industry today.

(CO2)